

Harry Hausner

Curriculum Vitae

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About Me

I am Harry, a neutrino experimentalist interested in exploring 4-flavor oscillation physics and physics beyond the standard model with accelerator neutrinos. I received my PhD from the University of Wisconsin–Madison where I worked in Brian Rebel's group, implementing a covariance matrix fitting technique for use in searches for sterile neutrinos at NOvA. Upon completion of my doctoral program I joined Fermilab as a postdoctoral researcher under Angela Fava. Here I have brought my experience with systematics to help push the ICARUS single-detector muon neutrino disappearance search and prepare for the SBN multi-detector search. This joint analysis will make a definitive measurement oscillations due to $e\nu^2$ -scale mass-squared splitting.

In addition to my research, I find it important to foster community within neutrino physics; especially as it relates to helping younger physicists find their footing. I take great pride in having severed multiple times as a SULI mentor, guiding undergraduate students in a research project; and have been actively involved in SBN Young, a grassroots organization of graduate students and other young researchers within the SBN Program where we can support and help one another. There is no physics without the physics community, and each of us has a role to play in helping each other achieve the best science possible.

Education & Work

- **Postdoctoral Research Associate** 2022–Present
Fermi National Accelerator Laboratory
- **PhD in Physics** August 2022
University of Wisconsin–Madison
- **MA in Physics** August 2019
University of Wisconsin–Madison
- **BS in Physics & Mathematics** *summa cum laude* June 2016
Union College – Schenectady, NY

Awards & Honors

- *Van Vleck Fellowship in Physics* 2016
University of Wisconsin–Madison
- *Sigma Pi Sigma Physics Honors Society* 2015
Union College – Schenectady, NY
- *Pi Mu Epsilon Math Honors Society* 2014
Union College – Schenectady, NY

Research

- **Detector Hardware** 2022-Present
 - Implemented TPC hardware compression algorithm needed to reduce data footprint and increase data acquisition throughput
 - Leading project to assess the use of Broadcom SiPMs in cryogenic conditions as part of APEX photon collection system for potential use in DUNE FD Phase 3
- **Neutrino Physics with ICARUS/SBN** 2022-Present
 - Investigated single photon shower reconstruction in ICARUS via machine learning techniques
 - Implement workflows with Fermilab's Elastic Analysis Facility to use Fermilab GPU resources for machine learning reconstruction for SBN physics
 - Convened ICARUS Systematics working group to organize our handling of systematic uncertainties for all analyses
 - Implemented detector data quality metrics used for all analyses to ensure only physics quality data is used for published ICARUS results
 - Implemented the WireMod tool from MicroBooNE to generate MC samples for detector systematic uncertainties using modified TPC waveforms
 - Helped implement HARPS (Hit Activity Removal from Particles For Systematics) tool to improve treatment of ICARUS systematic uncertainties
- **Data Acquisition and Detector Operations at ICARUS** 2022-Present
 - Served as ICARUS data acquisition expert to diagnose and debug issues with the data acquisition in real time during detector operations
 - Served as ICARUS Run Coordinator organizing detector upgrades and studies during beam shutdown and ensuring optimal detector functionality
- **Sterile Neutrino Searches with NOvA** 2018-2022
 - Validated NOvA Monte Carlo to check for errors in simulation production
 - Investigated alternative calibration methods for NOvA detectors to increase statistics used in data-driven energy estimation
 - Designed NOvA's neutral current event selection criteria to ensure well reconstructed events
 - Derived a method of estimating the calorimetric energy deposited in the NOvA detectors by neutral current neutrino interactions
 - Worked in the NOvA production group to produce the simulated neutrino events needed for all analyses performed with NOvA data
 - Analyzed data to measure potential sterile neutrino appearance via deficits in neutral current interactions and changes in muon survival probability
 - Helped implement a covariance matrix fitting framework to allow for a two detector fit of neutrino interactions observed by NOvA

Recent Talks & Publications

- Talk: "Single Photon Searches at ICARUS with SPINE", Neutrino Physics and Machine Learning, (October 29th, 2025)
- Talk: "Latest Results from the ICARUS Experiment at the Short-Baseline Neutrino Program", Conference on the Intersection of Particle and Nuclear Physics (June 9th, 2025)
- Publication: ICARUS Collaboration, "Search for a Hidden Sector Scalar from Kaon Decay in the Dimuon Final State at ICARUS", Phys.Rev.Lett. 134 (2025) 15, 151801
- Publication: NOvA Collaboration, "Dual-Baseline Search for Active-to-Sterile Neutrino Oscillations in NOvA", Phys.Rev.Lett. 134 (2025) 8, 081804

Teaching & Mentorship

- **Italian Summer Student Mentor** 2024
Fermi National Accelerator Laboratory
 - Mentored graduate student as part of a collaborative program between INFN in Italy and Fermilab in the United States
 - Gave guidance on research task and trained him to perform technical tasks
 - Helped student revise and write a report on his research
- **SULI Mentor** 2023-2024
Fermi National Accelerator Laboratory
 - Mentored undergraduate students as part of DOE Science Undergraduate Laboratory Internships program
 - Organized research tasks for students to help develop their skills as scientists
 - Helped students prepare to present posters on their research
- **Teaching Assistant** 2016-2018
University of Wisconsin–Madison Department of Physics
 - Assisted in teaching undergraduate mechanics at the introductory level
 - Lead discussion groups to help students personally understand the material
 - Designed assessments to measure students' retention of information

Organizational Experience

- **Neutrino Physics Center** 2025-Present
Fermi National Accelerator Laboratory
 - Coordinated the NPC journal club to engage the neutrino community in discussing current research and inviting guests to discuss their work
- **ICARUS Shift Czar** 2022-2023
ICARUS Collaboration
 - Coordinated shift assignment for the ICARUS collaboration to ensure smooth detector operations
- **Teaching Assistance Association Physics Department Steward** 2016-2017
University of Wisconsin–Madison
 - Organized and surveyed physics department graduate workers to improve working conditions
- **Society of Physics Students** 2012-2016
Union College – Schenectady, NY
 - Organized physics outreach events to educated community about the wonders of science